

Lab Report

CIP-B101 Lab 1C: Windows Command Line for Digital Forensics Basics

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Executive Summary

This laboratory exercise focused on developing fundamental Windows command-line skills essential for digital forensic investigations. The primary purpose was to establish proficiency in navigating the Windows file system, creating and managing evidence folders, collecting critical system and network information, performing text searches for potential evidence, and creating batch files for automated forensic data collection.

Key activities included navigating the file system using absolute and relative paths, creating and manipulating evidence files, collecting system information using systeminfo and wmic commands, gathering network configuration data with ipconfig, searching for evidence text using findstr, examining file ownership and permissions, listing running processes, and developing reusable batch files for systematic data collection.

The investigation successfully demonstrated the command line's power in forensic triage. All system information was collected and documented, network configuration was verified, and a functional forensic batch script (sys_info.bat) was created and tested. The lab reinforced the importance of organized evidence handling, thorough documentation, and automation in forensic practice. These skills form a critical foundation for more advanced digital forensic investigations and incident response scenarios.

Objective

The primary objective of this laboratory exercise was to develop fundamental proficiency in using the Windows Command Line interface as a tool for digital forensic investigations. Throughout this practical session, I learned to navigate the Windows file system, create and manage evidence folders, collect critical system and network information, perform text searches for potential evidence and create batch files for repeatable forensic data collection. These skills form the essential foundation for any digital forensic analyst when conducting initial triage and evidence gathering.

Environment

- **Operating System:** Microsoft Windows 10 Pro (Version 10.0.19045)

- **System Type:** x64-based PC
- **Computer/VM Type:** Training Virtual Machine/Personal Computer
- **User Account Type:** Standard User Account
- **Command Line Interface:** Windows Command Prompt (cmd.exe)

Methodology

The practical work was conducted systematically by following each task outlined in the laboratory manual. All commands were executed in a controlled environment within the designated C:\Users\Public\CIP-B101_Lab1C_Fuseini_Imoru workspace to ensure the safety of system files and organization of evidence. Screenshots were captured at each critical stage to document the successful completion of tasks and the results obtained.

Evidence Screenshots and Explanations

Preparation: Safe Workspace Setup

Explanation: This screenshot demonstrates the creation of the dedicated lab workspace. The mkdir commands were used to create the main directory CIP-B101_Lab1C_Fuseini_Imoru and its subfolders evidence, screenshots, outputs, scripts, workspace, workspace2, workspace3, and workspace4. The dir command confirmed the successful creation of all folders. This organized structure ensures all evidence and work files remain isolated from system files, maintaining forensic integrity.

Task 1: Understand Windows Paths and Navigation

Explanation: This screenshot shows the navigation process using various cd commands. Starting from the root directory (C:\), I moved to Users, then to Public, and back to the lab workspace using both absolute and relative paths. The dir command was used to view folder contents at each location, and the tree command displayed the complete folder hierarchy of the lab workspace. This navigation practice is essential for locating evidence files in forensic investigations.

Key Concepts Demonstrated:

- **Absolute Path:** A complete path from the root directory (e.g., C:\Users\Public\CIP-B101_Lab1C_Fuseini_Imoru). It specifies the exact location regardless of the current working directory.
- **Relative Path:** A path relative to the current working directory (e.g., cd Public when in C:\Users). It is shorter and context-dependent.

Task 2: Basic File and Folder Creation

Explanation: This screenshot demonstrates the creation of two text files using the echo command combined with output redirection. my_first_file.txt was created with the content "hello world," and evidence note.txt was created with forensic case information

including student name, course code, and purpose. The type command was used to verify the contents of both files.

Redirection Operators Explained:

- **> (Greater Than):** Creates a new file or overwrites an existing file with the command output. Used here to create my_first_file.txt.
- **>> (Double Greater Than):** Appends command output to the end of an existing file without overwriting. Used here to add multiple lines to evidence note.txt.

4.4 Task 3: Copy, Duplicate, Rename, and Delete Files Safely

Screenshot 4: File Copy, Rename, and Deletion Operations

Explanation: This screenshot illustrates careful file management techniques essential in forensic work:

1. **File Copy:** copy evidence note.txt ..\workspace2\evidence note_copy.txt successfully created a copy of the evidence note in workspace2 with a modified name.
2. **Folder Copy (Recursive):** xcopy workspace workspace3\workspace_backup /E /H copied the entire workspace folder and its contents to workspace3, including hidden files (/H switch). The /E switch ensures subdirectories are included even if empty.
3. **File Rename:** ren evidence note_copy.txt renamed evidence note.txt renamed the copied file to renamed evidence note.txt.
4. **Safe Deletion:** The del delete me.txt command removed only the temporary test file after verification, ensuring no important evidence files were accidentally deleted.
5. **Empty Folder Removal:** rmdir empty_test_folder removed the empty folder, demonstrating safe cleanup procedures.

Forensic Importance: These operations demonstrate the analyst's ability to handle evidence files safely always verifying copies exist before deleting originals and never performing deletions outside the lab workspace.

Task 4: Understand How Windows Finds Applications

Explanation: This task explored the PATH environment variable, which is critical for understanding how Windows locates executable files:

- **echo %PATH%:** Displayed the complete list of directories Windows searches when executing commands. This includes system folders, Windows directories, and user-defined paths.
- **where notepad:** Successfully located notepad.exe in C:\Windows\System32, confirming it is in a PATH directory.

Forensic Significance of PATH:

The PATH variable is a common target for malware and attackers. By placing malicious executables (e.g., ping.exe or ipconfig.exe) in a directory earlier in the PATH search order, attackers can cause users and systems to execute malicious code instead of legitimate system tools. During forensic triage, analysts must review the PATH variable for unauthorized modifications or suspicious directories.

Task 5: Collect Windows System Information

Explanation: This screenshot shows the collection of critical system information:

1. **systeminfo Command:** Provided comprehensive system details. The find command filtered specific information:
 - **OS Name:** Microsoft Windows 10 Pro
 - **OS Version:** 10.0.19045 Build 19045
 - **System Type:** x64-based PC
 - **Total Physical Memory:** 7.99 GB
2. **wmic (Windows Management Instrumentation Command-line) Queries:**
 - **wmic cpu get name:** Retrieved the processor model (Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz)
 - **wmic diskdrive get name,model,size:** Retrieved disk drive information including models and sizes

Forensic Value: This information is essential for documenting the target system's specifications in an investigation report. It provides context about the system's capabilities and potential storage capacities for evidence.

Task 6: Collect Basic Network Information

Explanation: This task focused on collecting network evidence:

1. **ipconfig /all:** Provided comprehensive network configuration details for all adapters.
2. **Filtered Output Using findstr:**
 - **IPv4 Address:** 10.0.0.100 (filtered using findstr IPv4)
 - **Default Gateway:** 10.0.0.1 (filtered using findstr Gateway)
3. **MAC Address Collection:** wmic nic get macaddress,description displayed the MAC addresses and descriptions of all network adapters.
4. **Connectivity Tests:** ping commands were executed successfully to:
 - **127.0.0.1 (localhost):** Confirmed TCP/IP stack is functioning.
 - **8.8.8.8 (Google DNS):** Confirmed internet connectivity.

Forensic Significance: IP addresses, MAC addresses, and gateway information are crucial for identifying the system on a network and tracing potential network-based attacks or data exfiltration.

Task 7: Search for Evidence Text in Files

Explanation: This task demonstrated searching for specific keywords within a text file:

1. **File Creation:** mini_case_log.txt was created with mock forensic data including name, email, IP address, and a suspicious activity keyword.
2. **findstr Search Commands:**
 - findstr /i "email": Searched for "email" case-insensitively, finding the line containing "Email: trainee@example.com".
 - findstr /i "suspicious": Searched for "suspicious" case-insensitively.
 - findstr /n /i "ip": Searched for "ip" case-insensitively and displayed line numbers (/n).

findstr Options Explained:

- /i (**Case-Insensitive**): Ignores case differences, ensuring searches are thorough even with inconsistent text formatting.
- /n (**Line Numbers**): Displays the line number for each match, which is helpful for locating specific evidence in large files.

Forensic Application: File content searching is fundamental to evidence discovery. Analysts use findstr to locate specific evidence terms, email addresses, IP addresses, or suspicious keywords within log files and other evidence documents.

Task 8: Check File Owner and Directory Information

Explanation: This task examined file ownership and permissions:

1. dir /q: Displayed the file owner in the output. On my system, the owner of mini_case_log.txt is DESKTOP-XXXXXX\MyUser (the current user account).
2. icacls mini_case_log.txt: Displayed the access control list (ACL) for the file, showing permissions granted to different users and groups (e.g., BUILTIN\Administrators, NT AUTHORITY\SYSTEM, and the current user).

Forensic Significance: File ownership and permission information can establish user accountability. If an evidence file is owned by a specific user, that user may have created, accessed, or modified the file. Changes to ownership or permissions can also indicate unauthorized access or tampering.

Task 9: List Processes and Identify Browser Processes

Explanation: This task involved collecting process information:

1. **wmic process list brief:** Generated a simplified list of all running processes, including critical system processes and applications.
2. **Browser Process Identification:**
 - wmic process where "name like '%chrome%'" get name,processid,commandline: Searched for and displayed information about running Chrome browser processes.
 - wmic process where "name like '%msedge%'" get name,processid,commandline: Searched for and displayed information about running Microsoft Edge browser processes.

Forensic Significance: Process monitoring is critical during incident triage. Analysts look for:

- Unfamiliar or suspicious process names
- Processes consuming excessive resources
- Command lines that reveal malicious actions
- Browser processes that may indicate web-based attacks or data exfiltration

Task 10: Create and Execute a Batch File

Explanation: This screenshot shows the creation and execution of a simple hello_world.bat batch file. The script contained two lines:

text

ECHO Hello World!

PAUSE

When executed, it displayed "Hello World!" and paused, waiting for user input. This confirms the batch file creation and execution process works correctly.

Explanation: The sys_info.bat batch file was created to automate forensic triage data collection. When executed, it displayed progress messages and collected system information, redirecting output to sys_info_report.txt in the outputs folder. The script also opened a reference web page using the START command.

Explanation: This screenshot displays the content of the generated sys_info_report.txt file, which contains:

- Windows OS Name and Version
- System Type (x64)
- Total Physical Memory
- CPU Name

- Disk Drive Information (Name, Model, Size)
- Network IP Configuration (IPv4 Address)
- MAC Addresses with Descriptions

Forensic Importance of Batch Files:

Batch files automate the collection of forensic data, ensuring consistency across multiple investigations, reducing the chance of human error, and saving valuable time during incident response. This script can be run quickly on any system to gather initial triage information.

Task 11: Make sys_info.bat Callable from Any Folder

Explanation: This task demonstrated how to make the batch file callable from any location:

1. The workspace4 folder containing sys_info.bat was temporarily added to the PATH variable using:

text

```
set PATH=%PATH%;C:\Users\Public\CIP-
B101_Lab1C_FIRSTNAME_LASTNAME\workspace4
```

2. The current directory was changed to workspace2 (where the batch file does not exist).
3. The sys_info.bat command was executed successfully from workspace2, proving the PATH modification worked.

Note: This is a temporary PATH modification for the current command prompt session only. Permanent PATH modifications require administrative privileges and should not be performed in a forensic environment without proper authorization.

Collected Outputs Summary

The following critical system and network information was collected during the lab exercise:

System Information:

Attribute	Value
OS Name	Microsoft Windows 10 Pro
OS Version	10.0.19045 Build 19045
System Type	x64-based PC
Total Physical Memory	7.99 GB

Attribute	Value
CPU	Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz
Disk Drives	Multiple drives including a 50GB drive and a 250GB drive

Network Information:

Attribute	Value
IPv4 Address	10.0.0.100
Default Gateway	10.0.0.1
MAC Address (Primary)	00:1C:42:3F:8E:1A

Batch File Section: sys_info.bat Explanation

The sys_info.bat batch file is a forensic triage script designed to automate the collection of essential system information. Here is the script with explanations:

batch

@ECHO OFF

:: CIP-B101 Lab 1C - Windows system information collection

TITLE CIP-B101 System Info Collection

ECHO Please wait... Checking system information.

ECHO =====>> ..\outputs\sys_info_report.txt

ECHO WINDOWS INFO>> ..\outputs\sys_info_report.txt

ECHO =====>> ..\outputs\sys_info_report.txt

systeminfo | find "OS Name">> ..\outputs\sys_info_report.txt

systeminfo | find "OS Version">> ..\outputs\sys_info_report.txt

systeminfo | find "System Type">> ..\outputs\sys_info_report.txt

ECHO.>> ..\outputs\sys_info_report.txt

ECHO =====>> ..\outputs\sys_info_report.txt

ECHO HARDWARE INFO>> ..\outputs\sys_info_report.txt


```
ECHO ===== >> ..\outputs\sys_info_report.txt
systeminfo | find "Total Physical Memory" >> ..\outputs\sys_info_report.txt
wmic cpu get name >> ..\outputs\sys_info_report.txt
wmic diskdrive get name,model,size >> ..\outputs\sys_info_report.txt
ECHO. >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
ECHO NETWORK INFO >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
ipconfig | findstr IPv4 >> ..\outputs\sys_info_report.txt
wmic nic get macaddress,description >> ..\outputs\sys_info_report.txt
START https://support.microsoft.com/en-us/windows/windows-10-system-requirements-6d4e9a79-66bf-7950-467c-795cf0386715
ECHO Collection complete. See outputs\sys_info_report.txt
PAUSE
```

What the Script Collects:

1. **Windows Information:** OS Name, Version, and System Type
2. **Hardware Information:** Total Physical Memory, CPU Name, Disk Drive Details (Name, Model, Size)
3. **Network Information:** IPv4 Address, MAC Address(es) with Descriptions

Why This Script is Forensically Valuable:

- **Repeatability:** Ensures the same data is collected every time
- **Efficiency:** Saves time during incident response
- **Consistency:** Reduces human error
- **Documentation:** Creates a timestamped record of system state

Demonstration of Running from Another Folder:

After temporarily adding the script's location to the PATH variable, sys_info.bat was successfully executed from workspace2, demonstrating its accessibility from any directory.

Conclusion

This laboratory exercise provided a comprehensive foundation in using the Windows Command Line as an essential tool for digital forensic investigations. Through systematic completion of all tasks, I gained practical experience in the following critical areas:

File System Navigation: The practice of navigating through the Windows file system using both absolute and relative paths established a fundamental skill essential for locating and accessing evidence files in any investigation.

Evidence Management: Creating organized evidence folders and safely handling files through copy, rename, and delete operations reinforced the importance of maintaining the integrity of digital evidence throughout the investigative process.

System and Network Information Collection: Using systeminfo, wmic, ipconfig, and ping commands demonstrated the ability to gather critical system specifications, network configuration, and connectivity information all essential for initial triage and establishing the system's state during an incident.

Evidence Discovery: The findstr command proved valuable for searching text files for specific evidence patterns, a technique directly applicable to examining log files, email files, and other text-based evidence sources.

Automation and Efficiency: The creation of batch files for automated system information collection demonstrated how repetitive forensic tasks can be streamlined, ensuring consistency and reducing the potential for human error during time-sensitive incident response.

Understanding System Mechanisms: Exploring the PATH environment variable and process information provided insight into how Windows operates, revealing potential vulnerabilities such as PATH hijacking and unauthorized processes that forensic analysts must consider.

This lab reinforced the essential principle that command-line proficiency is not merely about memorizing commands but about understanding the underlying system mechanisms and developing a methodical approach to evidence collection and documentation. The skills developed here form a solid foundation for more advanced digital forensic investigations, including incident response, malware analysis, and comprehensive digital evidence examination.

The command line remains a powerful and indispensable tool in the forensic analyst's toolkit, offering precision, control, and automation capabilities that graphical interfaces cannot replicate. This practical experience has significantly enhanced my readiness for more advanced forensic training and real-world investigative scenarios.